

ISS5096: Experiments and Causal Inference

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Course Duration: Feb 23 – Jun 15, 2026

Classroom: TSMC Bldg. R406

Class Time: Thursday, 14:20 pm- 17:20 pm

TA: TBA

COURSE DESCRIPTION:

This course introduces experimental and quasi-experimental methods for causal inference that are widely used in a broad array of domains such as marketing and information systems. The focus of the course is on delivering a breadth of substantive topics and methodological considerations that emerge in utilizing the identification-oriented methods. Throughout the course, we will discuss topics that are related to methods such as randomized controlled trials (RCT), difference-in-differences (DiD), matching methods such as propensity score matching (PSM) and coarsened exact matching (CEM), and more advanced topics such as regression discontinuity designs (RDD), double-debiased machine learning (DML), synthetic control methods (SCM), and its extensions such as synthetic difference-in-differences (synth. DiD). To foster learning experiences, students will review relevant research papers on each topic and be asked to actively engage in presentations and discussions about the nature of causation and alternative means of inferring causal relationships.

Students will also carry out a collaborative group project for which they will design an experiment and associated plan of analysis in an attempt to draw business insights. Leading tech companies such as Netflix, Microsoft, and Amazon are growingly paying attention to the importance of business experimentation and causal inference to enhance their decision making, and are hiring employees that are equipped with such analytical tools/mindsets (e.g., watch [a recent video](#) highlighting this by causal inference science team at Netflix). In line with this growing trend, this course also caters to those that are interested in joining the industry after graduation with one condition: a willingness to work hard on possibly unfamiliar materials.

COURSE GOALS:

- Learn how to determine which methods and results best support empirical inference questions.
- Be familiar with causal inference methods that are widely used for business analytics.
- Understand the trade-offs in the design, analysis, and reporting of field/quasi/natural experiment methods.

PREREQUISITES:

- **Math:** undergraduate-level probability and statistics; some experiences with regression analysis/econometrics will be helpful.
- **Programming:** Knowledge of statistical programming (e.g., R and Python)
- **Recommended prior coursework:**
 - ISS5066 Programming for Business Analytics (PBA) covers statistical inference, R programming, and an introduction to causal inference including potential outcomes

and difference-in-differences, all directly relevant to ECI. ECI goes considerably deeper, with more formal notation and careful assumption-based reasoning. Students entering with PBA will find the foundational concepts familiar, but should expect to engage more rigorously with the underlying theory. Qualified undergraduates and graduate students from other departments are welcome to join the class.

- ISS5077 Computational Statistics for Data Science (CSDS) covers hypothesis testing, regression, and applied statistical modeling and will provide a strong background for the statistical foundations covered in this course.
- ISS5090 Business Analytics with Machine Learning (BAML) covers supervised and unsupervised learning methods including trees, random forests, and neural networks and will provide helpful background for the causal ML topics covered in Weeks 14 - 15.
- **Related coursework in other departments:**
 - ECON3033 Econometrics I and ECON3034 Econometrics II — regression methodology and empirical research designs.
 - ECON5099 Causal Inference in Econometrics covers related topics.

TEXTBOOK:

Students can choose to purchase either the hard copy or the e-book for both textbooks 1 and 2, but please be prepared to have the textbooks ready before the beginning of the semester. If you choose to order online, be aware that it might take time for the textbooks to be delivered. So, if you plan to take the course, please make sure to place your order early enough so the books arrive in time.

1. **Mostly Harmless Econometrics: an empiricist's companion** by J. D. Angrist and Jorn-Steffen Pischke, Princeton University Press, 2008.
2. **Causal Inference for Statistics, Social, and Biomedical Sciences** by G. Imbens and D. Rubin, Cambridge University Press, 2015.
3. **The Effect: An Introduction to Research Design and Causality** by N. Huntington-Klein, CRC Press, 2021. (Opensource PDF version provided by the author [here](#).)

The following books are optional but may prove useful for additional coverage of some of the course topics.

- J. D. Angrist, and Jorn-Steffen Pischke. 2015. *Mastering 'metrics: The path from cause to effect*. Princeton University Press.
- Cunningham, Scott. *Causal Inference: The Mixtape*. 2021. Yale University Press.
- Hernán, Miguel A., and James M. Robins. 2020. *Causal inference: What If*. CRC Press. Taylor and Francis Group. (Opensource PDF version provided by the authors [here](#).)
- Morgan, Stephen L., and Christopher Winship. 2015. *Counterfactuals and Causal Inference, Second Edition*. Cambridge University Press.
- Pearl, Judea. 2000. *Causality: Models, Reasoning, and Inference*. New York: Cambridge University Press.
- Blackwell, M., *A User's Guide to Statistical Inference and Regression*. (Opensource PDF version provided by the authors [here](#).)
- Rosenbaum, Paul R. 2002. *Observational Studies, 2nd edition*. Springer-Verlag.
- Rosenbaum, Paul R. 2009. *Design of Observational Studies*. Springer Series in Statistics.
- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data, Second Edition*. MIT Press.
- Rubin, Donald. 2006. *Matched Sampling for Causal Effects*. Cambridge University Press.

- Manski, Charles F. 1995. *Identification Problems in the Social Sciences*. Cambridge: Harvard University Press.
- Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. 2025. *Applied Causal Inference Powered by ML and AI*. (Unpublished. Opensource PDF version provided the authors [here](#).)

SOFTWARE:

We use R (with RStudio) as the primary computing environment. Key packages include: **tidyverse**, **fixest**, **did**, **rdrobust**, **MatchIt**, **weightIt**, **grf**, **dagitty**, **ggdag**, **modelsummary**. Students comfortable with Stata or Python may use those for their research projects, but in-class demonstrations and code examples will be in R.

COURSE EVALUATION:

Grading Policy

<i>Research Project</i>	50%
Problem Sets	15%
The Effect Assignments	15%
Participation (incl. presentations)	10%
One-Page Summaries	5%
Attendance	5%
Total	100%

Course Requirements

1. **AI Use Policy*:** Students are permitted to utilize AI tools or Large Language Models (LLMs; e.g., ChatGPT, Claude, Gemini) responsibly as aids for brainstorming and generating initial drafts for assignments. However, it is imperative that the final submission predominantly reflects the student's understanding and personal input. To maintain academic integrity, students who choose to use AI tools must adhere to the following guidelines:
 - a. **Thorough Explanation:** Students must provide a detailed explanation of how the AI tool was used in the completion of the assignment.
 - b. **Original AI Responses:** Alongside their submission, students are required to submit the original responses generated by the AI tool.
 - c. **Critical Evaluation and Personal Input:** Students should critically evaluate the AI-generated content and clearly indicate the portions of the work that have been modified or expanded upon with their own insights and understanding.

- d. **Proper Attribution:** Proper attribution should be given to the AI-generated content used, clearly indicating the sections that are AI-generated.

** Remember, attempting to cheat the system by heavily relying on AI-generated content without substantial personal input is ultimately the student's loss, as it undermines the learning process and personal growth.

2. **Academic Honesty and Plagiarism*:** All work submitted for academic evaluation must be the student's own. The penalty for violation of academic integrity will result in a zero for that assignment for the first time. Subsequent violation(s) will result in a failing grade for the course. Plagiarism will also not be tolerated.

Academic dishonesty comprises of, but is not limited to, the following:

- Cheating: Copying from other students' quizzes and assignments or allowing others to copy from one's own.
- Plagiarism: Using other people's original work without giving appropriate credit or acknowledgment to the authors or sources.
- Self-plagiarism: Submitting a piece of work in more than one course without the explicit permission of the instructors involved.
- Misrepresentation of authorship: Submitting work as one's own, which has been prepared by or purchased from others.

Students will be asked to upload their submission materials to Turnitin.com, an online plagiarism checker, to ensure academic integrity. Read more about online submission on <http://learning.site.nthu.edu.tw/p/412-1319-7120.php?Lang=en>.

3. **Attendance:** All students are expected to attend every class. Please bring your own hard copy of the course materials, which will be distributed by the instructor before class. If you have any urgent reason to miss a class, you are still responsible for the materials covered during the class and are expected to complete the required work. Attendance will be taken on a regular basis and will count towards your participation score (5%). Class missing will cause about 1% loss of final grade. Students who miss a class should inform me or the TA prior to the class via email or phone call.
4. **Attitude/Participation:** In class, the most important thing for the students is to stay active and engaged about the topic being discussed. Positive contributions to class discussions will increase your score towards attitude. When we discuss a topic in class, effective discussions are only possible if everyone is well prepared. Please, be prepared to open and engage in discussions with your opinions and thoughts.
5. **Problem Sets and 'The Effect' Assignments:** Methods are tools, and it isn't very instructive to read a lot about hammers or watch someone else wield a hammer -- you need to get your hands on a hammer or two. In this course, you will have 1) problem sets, which will be a mix of conceptual questions, analytic problems, computer simulations, and data analysis that closely resemble what we discuss in lectures, and 2) assignments from our third textbook, *The Effect*. I encourage you to work in groups on the homework, but you always need to write your own solutions including your computer code. Also, it is hugely beneficial to attempt the problems sets on your own before working in groups. Assignments are to be submitted on the due date. *Late submissions will be penalized 1 percentage point of the assignment's weight per day* (e.g., for an assignment worth 7 % of the course grade, turning it in 3 days late caps the attainable score at 4%).

6. **Journal Article Review**

- a. **One-Page Summaries:** Before class starts, every student should submit 1-page summary of the assigned papers which will be discussed in class. A guideline will be distributed for students to use when summarizing the papers.
- b. **One-Page Summary Guideline:**
 - i. Choose 5 papers from the assigned reading list and write a one-page summary addressing the following:
 1. Main Question: What is the main question of the paper? Why do we care?
 2. Parameters of Interest: What model aspects or facts are being estimated?
 3. Data: What is the unit of observation? Where are the data from? How were they collected?
 4. Answers: What does the paper find? What are the main weaknesses? How could it be improved?
 5. Extensions: After the current paper, what could be done next in this general area?
 - ii. You may also discuss: one-sentence conclusion, institutional background, conceptual framework, or relevant literature. Submit as PDF via Teams.
- c. **Paper Introduction Presentations (Peer-Reviewed):** Throughout the semester, students will be reading academic publications/papers that utilize different causal inference methods. Students will be asked to present the papers they reviewed each week (reading list will be distributed) by taking turns. The contents to be necessarily included are the value of the topic and motivational story. Other audience should also read the paper prior to the class for discussion. The presentations will be peer-reviewed.

7. **Research Project:** In lieu of a final exam, this course requires students to write a short paper applying or extending the causal inference methods we learn in this class. It should be no longer than 20 double-spaced pages and focus on the research design, data, methodology, results, and analysis.

- a. **Progress Report & Project Presentation (Peer-Reviewed):** Students will generate a potential research idea based on what we will discuss and suggest how to collect data and what causal inference methods/models can be used to carry out the research. The final project presentation will be peer-reviewed.
- b. **Structured Peer Feedback:** Final presentations are accompanied by structured peer feedback. After each presentation, every student completes a feedback form covering four areas: Research Question, Identification Strategy, Threats & Limitations, and a Constructive Suggestion. Feedback forms are shared with presenters to support their final revisions. The quality and thoughtfulness of your feedback contributes to your Participation grade.

Here is a brief timetable for the projects:

- c. **By Week 2:** Find a collaborator or obtain permission from the instructor to work on an individual project.
- d. **By Week 5:** Submit a short (half-page) description of your proposed project and a feasible plan for carrying out the research.
- e. **By Week 10:** Submit a brief (no longer than 5 page) page memo of your main results, including tables, figures, and brief analysis. For methodological projects, this should include a description of the method and any analytical/simulation results. You will be

required to give feedback on another group's project, which will be counted toward the overall grade based on attentiveness and usefulness of the feedback provided.

- f. **By Week 15:** Submit your final version of the project.

COURSE SCHEDULE:

Week	Topics	Reading
1	Introduction and Potential Outcomes: - Neyman-Rubin causal model - Fundamental problem of causal inference (FPOCI) - Causal estimands - Post-treatment bias under the truncation by death problem <u>Group forming & Ice breaking activities.</u>	- Imbens & Rubin, Ch. 1 - Angrist & Pischke, Ch.1 - Holland, P. W. 1986. “Statistics and Causal Inference”. Journal of the American Statistical Association, Vol. 81, No. 396: 945-960.
2	Randomization Inference: - Randomized experiments - Fisher’s approach to inference, permutation tests - Sharp null, randomization distribution <i>Due: The Effect CH. 2 Research Questions</i> <u>Find a collaborator for the group project.</u>	- Imbens & Rubin, Ch. 5 (Skim Ch. 4 for some definitions.) - Rosenbaum, Paul R. 2002. <i>Observational Studies</i>. Springer-Verlag. 2nd edition. Ch. 2
3	Inference for the Average Treatment Effect: - Neyman’s approach to inference for the ATE - Finite-sample vs superpopulation inference <i>Due: Problem Set 1 (Potential Outcomes); The Effect CH. 3 Describing Variables</i>	- Imbens & Rubin, Ch. 6, 9 (Skip 9.6–9.7), and 10 (Skip 10.6–10.7) - Angrist & Pischke: Ch. 2
4	Linear Regression and Randomized Experiments: - Simple linear regression in experiments - Covariate adjustment in experiments with regression <u>Students’ Paper Introduction Presentations: 1.1. and 1.2.</u> <i>Due: Problem Set 2 (Randomization Inference); The Effect CH. 4 Describing Relationships</i>	- Imbens & Rubin, Ch. 7, 9 (9.6–9.7), and 10 (10.6–10.7) - Winston Lin. “Agnostic notes on regression adjustments to experimental data: Reexamining Freedman’s critique.” Ann. Appl. Stat. 7 (1) 295 - 318, March 2013. - Freedman, David. 2008. “On regression adjustments to experimental data.”
5	Observational Studies I: - Selection on observables - Regression for observational data <u>Students’ Paper Introduction Presentations: 1.3. and 1.4.</u> <i>Due: Problem Set 3 (Inference for the ATE); The Effect CH. 13 Regression; Project Proposal</i>	- Angrist & Pischke, Ch. 3

6	Observational Studies II: - DAGs and the back-door criterion - Sensitivity analysis - Partial identification <u>Construct a DAG for your own research project!</u>	- Imbens & Rubin, Ch. 21 and 22 - Morgan & Winship, Ch. 4
7	In-Class Group Project Meetings (30 minutes per group): The purpose of the meetings is to help each group materialize their ideas/topics for the group project. <i>Due: The Effect CH. 6 Causal Diagrams, CH. 7 Drawing Causal Diagrams</i>	
8	Instrumental Variables I: - Noncompliance in randomized experiments - Instrumental variables in observational studies <u>Students' Paper Introduction Presentations: 2.1. and 2.2.</u> <i>Due: Problem Set 4 (Observational Studies); The Effect CH. 8 Causal Paths and Closing Back Doors</i>	- Imbens & Rubin, Ch. 23 and 24 - Angrist & Pischke, Ch. 4
9	Instrumental Variables II: - Two-stage least squares - Review of IV applications <u>Students' Paper Introduction Presentations: 2.3. and 2.4.</u>	
10	Panel Data, Fixed Effects, and Difference-in-Differences: - Fixed effects and first differences - Difference-in-differences <u>Students' Paper Introduction Presentations: 3.1. and 3.2.</u> <i>Due: Problem Set 5 (Instrumental Variables); The Effect CH. 19 Instrumental Variables; Progress Report</i>	- Angrist & Pischke, Ch. 5 - Imai, Kosuke and In Song Kim. (2019). When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data? American Journal of Political Science, Vol. 63, No. 2 (April), pp. 467-490.
11	Difference-in-Differences Cont.: - Applications of difference-in-differences <u>Students' Paper Introduction Presentations: 3.3. and 3.4.</u> <i>Due: The Effect CH. 18 Difference-in-Differences</i>	

12	Matching and Weighting Estimators: - Propensity scores, matching, and weighting <u>Students' Paper Introduction Presentations: 4.1., 4.2., 4.3., and 4.4.</u>	- Imbens & Rubin, Ch.13, 15, and 18 - Stuart, Elizabeth. 2010. Matching Methods for Causal Inference: A Review and a Look Forward. Stat. Sci. Vol. 25, No. 1: 1–21
13	Regression Discontinuity Designs: - Sharp RD designs, identification - Estimation and bandwidth selection <u>Students' Paper Introduction Presentations: 5.1. and 5.2.</u> <i>Due: The Effect CH. 14 Matching</i>	- Angrist & Pischke, Ch. 6 - Imbens, Guido W., and Thomas Lemieux. 2008. Regression Discontinuity Designs: A Guide to Practice. J. Econom. 142: 615-35.
14	Regression Discontinuity Designs Cont.: - Fuzzy RD designs. - Applications of RDDs. <u>Students' Paper Introduction Presentations: 5.3. and 5.4.</u> <i>Due: Problem Set 6 (Matching and Weighting Estimators)</i>	- Cattaneo, M., Idrobo, N., & Titiunik, R. (2020). A Practical Introduction to Regression Discontinuity Designs: Foundations (Elements in Quantitative and Computational Methods for the Social Sciences). Cambridge: Cambridge University Press. doi:10.1017/9781108684606 / Preprint Version
15	Advanced Topics (TENTATIVE) - One of the following topics will be covered (to be announced): i. <i>Option A</i> Causal Mechanisms: Mediation analysis, controlled direct effects, natural direct and indirect effects ii. <i>Option B</i> Causal Machine Learning: Heterogeneous treatment effects, CATE estimation, causal forests, double/debiased ML <i>Due: The Effect CH. 20 Regression Discontinuity; Final Report</i>	- Imai, K., Keele, L., & Yamamoto, T. (2010). Identification, inference and sensitivity analysis for causal mediation effects. <i>Statistical Science</i>, 25(1):51–71. - Imai, K., Keele, L., Tingley, D., & Yamamoto, T. (2011). Unpacking the black box of causality: Learning about causal mechanisms from experimental and observational studies. <i>American Political Science Review</i>, 105(4), 765-789. - Acharya, A., Blackwell, M., & Sen, M. (2016). Explaining causal findings without bias: Detecting and assessing direct effects. <i>American Political Science Review</i>, 110(3), 512-529.
16	Final Group Project Presentations	

READING ASSIGNMENTS:

Students will present papers from the following five topic groups. Each group contains five readings — a textbook chapter plus published applications from top journals. Presentation assignments will be finalized before the first presentation week.

Note: If you have a top-journal paper that uses a method covered in one of these modules (e.g., a DiD paper for the DiD module) and you would prefer to present it instead, please contact the instructor before presentations begin to request a substitution.

1. Reading Materials for Field Experiments:

- 1) *Chapter 2. The Experimental Ideal* in Joshua D. Angrist & Jörn-Steffen Pischke, 2009. "Mostly Harmless Econometrics: An Empiricist's Companion," Economics Books, Princeton University Press, edition 1, number 8769.
- 2) Elisa Montaguti, Scott A. Neslin, Sara Valentini (2016) Can Marketing Campaigns Induce Multichannel Buying and More Profitable Customers? A Field Experiment. *Marketing Science* 35(2):201-217.
- 3) Navdeep S. Sahni, Dan Zou, Pradeep K. Chintagunta (2017) Do Targeted Discount Offers Serve as Advertising? Evidence from 70 Field Experiments. *Management Science* 63(8):2688-2705.
- 4) Bapna, Ravi, Jui Ramaprasad, Galit Shmueli, and Akhmed Umyarov. "One-way mirrors in online dating: A randomized field experiment." *Management Science* 62, no. 11 (2016): 3100-3122.
- 5) *Chapter 1. Experiments and Generalized Causal Inference* in Cook, Thomas D., Donald Thomas Campbell, and William Shadish. *Experimental and quasi-experimental designs for generalized causal inference*. Boston, MA: Houghton Mifflin, 2002.

2. Reading Materials for Instrumental Variables (IVs):

- 1) *Chapter 4. Instrumental Variables in Action: Sometimes You Get What You Need* in Joshua D. Angrist & Jörn-Steffen Pischke, 2009. "Mostly Harmless Econometrics: An Empiricist's Companion," Economics Books, Princeton University Press, edition 1, number 8769.
- 2) Angrist, Imbens, G. W., & Rubin, D. B. (1996). Identification of Causal Effects Using Instrumental Variables. *Journal of the American Statistical Association*, 91(434), 444–455.
- 3) Florian Zettelmeyer, Fiona Scott Morton, & Jorge Silva-Risso. (2006). How the Internet Lowers Prices: Evidence from Matched Survey and Automobile Transaction Data. *Journal of Marketing Research*, 43(2), 168–181.
- 4) Dewan S, Ramaprasad J (2012) Research Note—Music Blogging, Online Sampling, and the Long Tail. *Inf. Syst. Res.* 23(3-part-2):1056–1067.
- 5) Barron, Kyle, Edward Kung, and Davide Proserpio. "The effect of home-sharing on house prices and rents: Evidence from Airbnb." *Marketing Science* 40.1 (2021): 23-47.

3. Reading Materials for Difference in Differences (DiD):

- 1) *Chapter 5. Parallel Worlds: Fixed Effects, Difference-in-Differences, and Panel Data* in Joshua D. Angrist & Jörn-Steffen Pischke, 2009. "Mostly Harmless Econometrics: An Empiricist's Companion," Economics Books, Princeton University Press, edition 1, number 8769.
- 2) Bertrand, Marianne, Esther Dufo, and Sendhil Mullainathan. "How much should we trust differences-in-differences estimates?" *The Quarterly journal of economics* 119.1 (2004): 249-275.
- 3) Dranove, David, et al. "Is more information better? The effects of "report cards" on health care providers." *Journal of political Economy* 111.3 (2003): 555-588.
- 4) Goldfarb, Avi, and Catherine E. Tucker. "Conducting research with quasi-experiments: A guide for marketers." *Rotman School of Management Working Paper* 2420920 (2014).
- 5) Foerderer, Jens, Nele Lueker, and Armin Heinzl. "And the Winner Is...? The Desirable and Undesirable Effects of Platform Awards." *Information Systems Research* 32, no. 4 (2021): 1155-1172.

4. Reading Materials for Matching Methods:

- 1) Gordon, Brett R., et al. "A comparison of approaches to advertising measurement: Evidence from big field experiments at Facebook." *Marketing Science* 38.2 (2019): 193-225.
- 2) Xu, Kaiquan, Jason Chan, Anindya Ghose, and Sang Pil Han. "Battle of the channels: The impact of tablets on digital commerce." *Management Science* 63, no. 5 (2017): 1469-1492.
- 3) Adamopoulos, Panagiotis, Vilma Todri, and Anindya Ghose. "Demand effects of the Internet-of-things sales channel: Evidence from automating the purchase process." *Information Systems Research* 32.1 (2020): 238-267.
- 4) Kim, Jun Hyung, et al. "Home-tutoring services assisted with technology: Investigating the role of artificial intelligence using a randomized field experiment." *Journal of Marketing Research* (2021): 00222437211050351.
- 5) Son, Yoonseock, Wonseok Oh, Sang Pil Han, and Sungho Park. "When loyalty goes mobile: Effects of mobile loyalty apps on purchase, redemption, and competition." *Information Systems Research* 31, no. 3 (2020): 835-847.

5. Reading Materials for Regression Discontinuity Design (RDD):

- 1) *Chapter 6. Getting a Little Jumpy: Regression Discontinuity Designs* in Joshua D. Angrist & Jörn-Steffen Pischke, 2009. "Mostly Harmless Econometrics: An Empiricist's Companion," Economics Books, Princeton University Press, edition 1, number 8769.
- 2) Caroline Flammer (2015) Does Corporate Social Responsibility Lead to Superior Financial Performance? A Regression Discontinuity Approach. *Management Science* 61(11):2549-2568.
- 3) Wesley Hartmann, Harikesh S. Nair, Sridhar Narayanan, (2011) Identifying Causal Marketing Mix Effects Using a Regression Discontinuity Design. *Marketing Science* 30(6):1079-1097.
- 4) Jo, Wooyong, et al. "Protecting consumers from themselves: Assessing consequences of usage restriction laws on online game usage and spending." *Marketing Science* 39.1 (2020): 117-133.
- 5) Flammer, Caroline, and Pratima Bansal. "Does a long-term orientation create value? Evidence from a regression discontinuity." *Strategic Management Journal* 38, no. 9 (2017): 1827-1847.

ADDITIONAL READING MATERIALS:

- Holland, Paul W. "Statistics and causal inference." *Journal of the American statistical Association* 81, no. 396 (1986): 945-960.
- Lin, Winston. "Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique." *The Annals of Applied Statistics* 7, no. 1 (2013): 295-318.
- Freedman, David A. "On regression adjustments to experimental data." *Advances in Applied Mathematics* 40, no. 2 (2008): 180-193.
- Stuart, Elizabeth A. "Matching methods for causal inference: A review and a look forward." *Statistical science: a review journal of the Institute of Mathematical Statistics* 25, no. 1 (2010): 1-21.
- Imai, Kosuke, and In Song Kim. "When should we use unit fixed effects regression models for causal inference with longitudinal data?" *American Journal of Political Science* 63, no. 2 (2019): 467-490.
- Baker, Andrew C., David F. Larcker, and Charles CY Wang. "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics* 144, no. 2 (2022): 370-395.
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review*, 111 (12): 4088–4118.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, James Robins, Double/debiased machine learning for treatment and structural parameters, *The Econometrics Journal*, Volume 21, Issue 1, 1 February 2018, Pages C1–C68, <https://doi.org/10.1111/ectj.12097>
- Angrist, Joshua D., and Jörn-Steffen Pischke. 2017. "Undergraduate Econometrics Instruction: Through Our Classes, Darkly." *Journal of Economic Perspectives* 31 (2): 125–44.
- Angrist, Joshua D., and Jörn-Steffen Pischke. 2010. "The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con out of Econometrics." *Journal of Economic Perspectives* 24 (2): 3–30.